

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Validating an Interpretable Machine Learning Framework for Emergency Department Triage at Mercer General: A Caribbean Context

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Statement of the Problem:

Emergency departments globally face persistent challenges of overcrowding, mistriage, and inefficient resource allocation, directly worsening patient outcomes and straining limited clinical resources (*Ariss et al., 2024; Chen et al., 2023; Xiao et al., 2023*). While machine learning models have demonstrated strong predictive performance for triage disposition, ICU admission, wait times and length of stay (*Chang et al., 2024; Pandey et al., 2024; Song et al., 2025; Wang et al., 2025*), all were validated exclusively in Western, East Asian, and Australian contexts, leaving Caribbean and low-resource emergency settings unrepresented. This study proposes validating an interpretable ML triage framework at Mercer General, addressing generalisability and health equity gaps identified across the reviewed literature.

Previous Work:

Substantial research over the past decade has applied machine learning to emergency department operations with promising results. Ariss et al. (2024) developed 144 voting ensemble models using the MIMIC-IV database to predict the full spectrum of ED resource needs, achieving an average AUC of 0.82, demonstrating that data-driven tools can meaningfully support triage resource allocation(1). Chang et al. (2024) integrated structured triage data with NLP-processed free-text nursing notes to predict patient disposition in Taiwanese EDs, with a Random Forest model outperforming emergency physicians on F1 score, demonstrating that combining structured and unstructured data significantly improves predictive performance(2).

For ICU-specific prediction, Chen et al. (2023) conducted a systematic review confirming that ML algorithms including Logistic Regression, Gradient Boosting, and Random Forest achieved AUC values ranging from 0.68 to 0.97 in identifying critically ill ED patients(3), while Pandey et al. (2024) developed an early ICU admission prediction model achieving an AUROC of 0.92, identifying triage category, ambulance arrival, and qSOFA score as the strongest predictors at the 30-minute mark (4). Complementing these findings, Lee et al. (2021) demonstrated that a neural network model trained on just eleven triage-available variables could predict hospitalization for urgent level-3 ED patients with a validation AUC

of 0.80, highlighting the viability of parsimonious models built on readily accessible clinical data (5).

Regarding operational metrics, Wang et al. (2025) demonstrated acceptable ML performance in predicting prolonged ED wait times, identifying ED crowding and mode of arrival as top predictors (6), while Song et al. (2025) developed interpretable models predicting both length of stay and disposition decision with accuracies exceeding 60% across ternary classifications (7). Building on the importance of rigorous data preprocessing, Naemi et al. (2021) quantified the impact of addressing missing values and data skewness in LOS prediction, demonstrating that applying multivariate imputation and resampling techniques improved ML model performance by an average of 15.75% in AUC and 32.19% in R^2 score, underscoring that data quality challenges must be explicitly addressed before deploying predictive models in clinical settings (8). Xiao et al. (2023) further advanced the field by proposing TransNet and TextRNN models that reduced under-triage by 12.06% and over-triage by 17.92% compared to manual triage (9).

However, real-world deployment frequently exposes a sharp divide between commercial or technical model capabilities and clinical reality. Wong et al. (2021) conducted an external validation of a widely deployed, proprietary clinical decision tool, the Epic Sepsis Model, and revealed poor real-world clinical performance, showing a low sensitivity of 33% that misses the vast majority of critical cases while generating a high volume of false alarms that actively exacerbate clinical alert fatigue (10). Furthermore, when these technologies are applied to high-stakes environments like triage and resource allocation, Cha and Kim (2025) noted that they introduce profound institutional vulnerabilities—such as data privacy breaches, algorithmic bias, and automation dependency—that highlight the urgent necessity of building human-centric, explainable AI architectures (11).

While predictive modelling studies have demonstrated strong technical performance, research from the Caribbean has highlighted the broader operational factors influencing emergency department efficiency. De Freitas et al. (2020) conducted a qualitative exploration of patient flow in a tertiary emergency department in Trinidad and Tobago and identified six key categories affecting patient movement, including organisational work processes, ED design and layout, material resources, staffing, non-clinical personnel, and external departments (12). Their findings suggest that successful optimisation of patient flow requires consideration of local workflow and organisational context in addition to predictive accuracy.

Beyond model development, successful implementation of clinical decision support systems remains a major challenge. Abell et al. (2023) reviewed 44 studies and identified 227 barriers and 130 facilitators affecting computerized clinical decision support system adoption, with workflow integration, output usefulness, and user trust emerging as the most

important determinants of successful implementation (13). Similarly, Giebel et al. (2025) reported that experts across healthcare, technology, legal, and regulatory domains emphasized the need for user involvement, high-quality datasets, mitigation of automation bias, and clear governance structures to support the safe integration of AI-based clinical decision support systems into clinical practice (14).

These governance and policy challenges are especially pronounced in low- and middle-income healthcare settings where complex data integrations occur. Mwogosi (2025) argued that localized healthcare frameworks often lack the resource strategies and specific regulatory guardrails needed to mitigate severe ethical risks like data re-identification and human oversight gaps during generative AI integration into electronic health records (15). To prevent structural inequalities and ensure clinical safety when deploying advanced medical algorithms, Taddese et al. (2025) emphasized that health systems must mandate meticulous data stewardship, strict quality management, and advanced cryptographic security to handle the complex, heterogeneous datasets underlying AI-driven applications (16).

Concerns regarding model transferability have also emerged. Yang et al. (2024) evaluated a COVID-19 triage AI model developed in United Kingdom hospitals and found that direct deployment in Vietnamese ICUs resulted in poor and highly variable performance, whereas transfer learning using local data substantially improved model accuracy (17). These findings demonstrate that AI models developed in high-income settings may not generalize effectively to low- and middle-income healthcare environments without local adaptation and validation.

Across all reviewed studies, however, predictive models were developed and validated predominantly in high-income, non-Caribbean settings, and none combined emergency department operational knowledge from Caribbean healthcare systems with machine learning-based triage prediction. Furthermore, fairness and health equity evaluations across demographic subgroups remain largely absent, while evidence on implementation readiness, workflow integration, and local model generalizability in Caribbean emergency departments is limited.

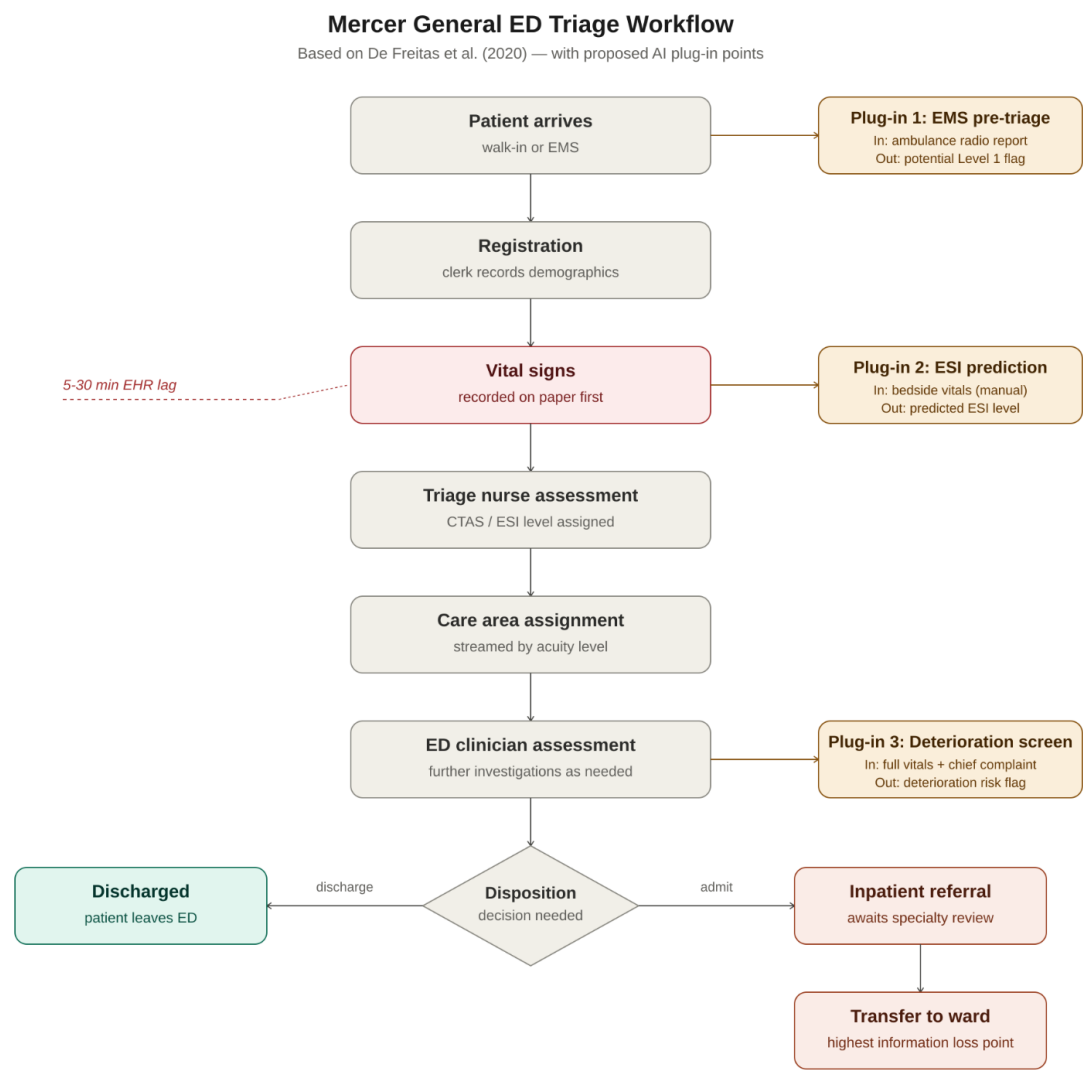
Proposed Solution:

This study proposes the development and validation of an interpretable machine learning framework tailored to the ED environment at Mercer General. Retrospective ED visit data will be extracted from Mercer General's electronic medical record system, incorporating structured triage variables; including vital signs, triage acuity, mode of arrival, demographics, and comorbidities, alongside unstructured free-text chief complaint notes processed through natural language processing.

Multiple ML algorithms will be trained and compared, including Random Forest, XGBoost, and Logistic Regression, with model interpretability ensured through SHAP values and feature importance analysis. The primary prediction targets will be clinical disposition, ICU admission likelihood, and prolonged wait time, evaluated using AUROC, F1 score, and Brier score.

Critically, this study will extend beyond aggregate performance metrics by conducting a subgroup fairness evaluation across race, ethnicity, language, and socioeconomic variables, directly addressing the health equity gap absent from all reviewed studies. The transparent framework will be designed for institutional adaptability, consistent with the approach recommended by *Song et al. (2025)*, enabling other Caribbean and low-resource hospitals to replicate the methodology using their own datasets.

Workflow Diagram:



Legend

Existing workflow step (De Freitas et al. 2020)

AI plug-in point (proposed)

Discharge outcome

Admission / transfer pathway

Documented friction point (constraint candidate)

Notes

Streaming and triage are combined at intake; patients are allocated to acuity-based streams immediately after assessment, consistent with the primary study's findings.

The transfer-to-ward step is the single largest source of delay and information loss in the entire flow, driven by specialty team review time and lack of inpatient beds.

Workflow Diagram Explanatory Notes

What the diagram shows

The diagram traces a patient's path through Mercer General's ED from arrival to disposition, in six main steps: *arrival* → *registration* → *vital signs* → *triage nurse assessment* → *care area assignment* → *ED clinician assessment* → *final disposition (discharge or admission/transfer)*. This is built directly from De Freitas et al.'s (2020)(12) observational study of a real Caribbean ED, which is the closest available evidence to Mercer's own context. That matters because the paper documents workarounds, not just the textbook process; e.g. vitals being recorded on paper before they ever reach the EHR.

Why these particular steps

Each box represents either a clinical decision point or a handoff between people, the two places where things tend to go wrong or slow down. Between steps, two friction points are explicitly marked because they shape where AI can realistically help:

1. The EHR lag (between vitals and triage assessment): vitals are taken on paper first and only entered into the EHR 5–30 minutes later. Any AI tool that assumes it can read "live" EHR data at this point would actually be working from stale or missing information.

2. The handover/transfer point (after a disposition to admit): De Freitas et al. found this is where the *most* information gets lost in the entire process, since it relies on verbal handoffs between ED and inpatient staff.

Why the AI plug-in points are placed where they are

Each plug-in is positioned based on *when the data it needs actually exists*, not just where it would be conceptually convenient:

- **Plug-in 1 (EMS pre-triage flag)** sits before the patient even arrives, using the ambulance radio call, so the nurse gets a heads-up before the patient is physically in the building.

- **Plug-in 2 (ESI prediction)** sits right after vitals are taken, but because of the EHR lag constraint, it has to pull from a manual-entry interface fed by the nurse directly, not from the EHR.

- **Plug-in 3 (deterioration screen)** sits at the end of clinician assessment, once full vitals and chief complaint are both available, to flag risk before a disposition decision is finalized.

The overall point of the diagram

It's meant to show that the AI isn't just "added on top" of the workflow. Instead, it is threaded into specific moments where real data exists and a human is already making a decision. The diagram also makes the constraints visible rather than abstract: you can see exactly where the EHR lag and handover loss happen, which is what justifies the three constraints documented separately.

Constraints:

The constraints below were tested against the preference-versus-constraint criterion: if ignoring the item causes the system to fail or produce harm, it is a constraint; if the system still works without it, it belongs on a wish list instead. One constraint is documented per category: technical, operational, and structural; consistent with the finding that most clinical AI deployments fail on the latter two, not the former.

Constraint 1: EHR Data Lag (Technical)

- **Mechanism:** Vitals are recorded on paper at the bedside first and entered into the EHR later, introducing a documented 5–30 minute delay before structured data exists in the system (De Freitas et al., 2020).
- **Design impact:** Plug-in 2 (ESI prediction) cannot rely on EHR-pulled vitals at the point of assessment; the data simply is not there yet.
- **Mitigation:** Provide a lightweight manual-entry interface so the nurse's bedside reading feeds the model directly; treat EHR-sourced vitals as a fallback only, not the primary input.

Constraint 2: Triage Nurse Time Pressure (Operational)

- **Mechanism:** A triage nurse assesses 15–30 patients per shift, frequently under-staffed. De Freitas et al. (2020) documented night shifts running with as few as 4 nurses against a target of 14.
- **Design impact:** Any plug-in requiring an extra click, manual override, or mandatory screen check per patient is a real time and cognitive cost that competes directly with bedside care.
- **Mitigation:** Default all AI output to passive display, visible alongside the nurse's own assessment, rather than an active step the nurse must clear before proceeding.

Constraint 3: Handover Information Loss (Structural)

- **Mechanism:** De Freitas et al. (2020) identified shift handover as the highest information-loss point in the entire ED flow, with verbal handoffs and no structured capture of flagged risk.

- **Design impact:** A deterioration flag raised on one shift (Plug-in 3) can silently disappear at handover if it exists only as a dashboard alert and is not written into a field the incoming nurse will see.

- **Mitigation:** Persist any AI-generated flag as a visible, timestamped field in the patient's chart, so it survives handover regardless of what is communicated verbally.

Stakeholders:

Stakeholders	Primary Concern	What Success Looks Like
Triage Nurse	Accuracy & Speed - <i>a wrong triage assignment delaying a Level 1 patient by even a few minutes can cause harm</i>	Overrides take one glance and 1 tap; she trusts her own judgements instead of fighting the system.
ED Consultant	Clinical defensibility - <i>a patient's outcome is questioned in a review, and the consultant needs to be able to show what exactly the model predicted and why</i>	Can pull a clear record of the model inputs, and predictions for any patient almost immediately.
Porter/ Support	Workflow disruption - <i>if an AI flag changes who urgently needs to be moved, but that change isn't communicated the usual way (like verbal), he could move the wrong patient in the wrong timing</i>	Urgent moves will still come through the normal verbal system, and no new screen or step has to be added to his job
Admin Clerk	Data entry burden - <i>if a patient arrives without ID and cannot confirm their date of birth, they still have to be registered, but if the model requires certain data before running, there can be a huge bottleneck in patient recordings.</i>	The model can run with partial registration without blocking the rest of patient inflow.
Patient Administrator	Equity & transparency - <i>if the model is trained on mostly majority-demographic data, it could be less accurate for older, non-English-speaking patients.</i>	A published fairness check exists showing no significant accuracy gap across demographic groups.

Risk Analysis:

Risk Register – Mercer General AI Triage System

Project: AI-Assisted ED Triage Validation Pilot

Institution: Mercer General Hospital

Prepared for: Saint Cedric Ministry of Health

Register Structure

Each row contains six fields:

- 1. **Risk Name** - Short, specific label
- 2. **Category** - AI-Technical · Operational · Ethical · Equity
- 3. **Likelihood** - L / M / H with clinical justification
- 4. **Impact** - L / M / H with clinical consequence
- 5. **Mitigation** - Specific design or process change (Owner + Frequency + Threshold + Trigger)
- 6. **Signal of Success** - Measurable indicator that the mitigation is working

	Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
1	Distribution Shift	AI-Technical	H — Mercer data covers limited seasonal and outbreak patterns (e.g. dengue, carnival surges) not represented in training	H — Model misclassifies critically ill patients during atypical presentation periods, delaying ICU referral within first 30 minutes	Clinical AI Lead re-evaluates model recall monthly against a 10% holdout; a >3 pp drop in Level 1 recall triggers mandatory clinical review before continued use	Monthly recall report published; zero unreviewed drops >3 pp in past quarter
2	Overfitting to Single-Site Data	AI-Technical	H — Model trained exclusively on Mercer General; no external Caribbean validation cohort exists yet	H — Model fails silently at other Saint Cedric public hospitals with different patient demographics and ED workflows	External validation cohort from at least one additional Caribbean hospital must achieve recall within ±3 pp of Mercer training-site recall before any nationwide rollout	Pre-rollout validation report signed off by Dr. De Freitas and Ministry reviewer
3	Data Leakage	AI-Technical	M — Post-triage variables (e.g. LOS, final disposition status) may have been inadvertently included during feature engineering	H — Model performs well offline but fails in real-time deployment, causing overtriage or undertriage of ESI level 3–5 patients within the first shift	Pre-deployment feature audit by a data engineer independent of model development; only variables available at the 30-minute triage mark permitted as inputs	Feature audit checklist signed and committed to repo; zero post-triage variables confirmed in production pipeline

4	Data Quality and Missingness	AI-Technical	H — Caribbean EHR systems often have incomplete vital sign records, particularly oxygen saturation and weight	M — Missing features cause the model to default to unreliable imputed values, degrading triage accuracy for walk-in patients without ambulance records	Multivariate imputation strategy documented and validated; model retrained quarterly incorporating missingness flags as explicit features; missing data rate per variable reported monthly	Monthly missingness dashboard reviewed; no input variable exceeds 10% missing rate in production without triggering retraining review
5	AI Plug-in Timing Mismatch	AI-Technical	H — Vitals recorded on paper before EHR entry (5–30 min lag); Plug-in 2 cannot reliably pull live EHR data at the moment of triage nurse assessment	H — If model is triggered before required input data exists, it runs on stale or null values and produces an unreliable prediction, or fails silently	Each plug-in assigned a defined data-readiness trigger; Plug-in 2 activates only after manual bedside vital entry into model interface, not from EHR pull; validated in pre-deployment simulation using 500 historical records	Pre-deployment simulation confirms each plug-in fires only when required inputs are present in ≥95% of test cases; zero null-input predictions in first 30 days of pilot
6	Alert Fatigue	Operational	H — High ED volume at Mercer General means frequent low-confidence alerts will quickly erode clinician attention	H — Triage nurses begin ignoring model alerts within weeks of deployment, causing undertriage of ESI level 2 patients	Alert threshold set at ≥75% predicted probability for ICU referral triggers, reviewed and adjusted by Clinical AI Lead every 30 days based on alert action rate	Percentage of alerts triggering a documented clinical action monitored weekly; rate below 40% for two consecutive weeks triggers threshold review
7	Clinician Over-Reliance	Operational	M — Triage nurses with limited AI experience may defer entirely to model output without independent assessment	H — A critically ill elderly patient with atypical symptoms is undertriaged and deteriorates in the waiting area because model error goes unchallenged	Human-in-the-loop design enforced: model output displayed as recommendation only; clinician must record their own assessment before seeing model score in first 90 days	Override rate monitored weekly; sudden drop below 5% overrides triggers mandatory simulation drill and departmental review
8	EHR Data Incompleteness During Retrospective Extraction	Operational	H — De Freitas et al. (2020) documented vitals recorded on paper first, entered into EHR 5–30 minutes later; retrospective records may contain missing, delayed, or inconsistently formatted entries	H — Incomplete or corrupted retrospective records reduce usable training data, introduce systematic bias, and may cause the model to perform well on clean records but fail on incomplete records most common in real use	Data completeness audit conducted by lead data engineer before model training; variables with >10% missingness flagged for multivariate imputation; records with >30% missing fields excluded with documented justification	Data completeness audit report committed to repo before model training begins; missingness rate per variable published in the methods section of the final report
9	Automation Bias	Ethical	H — Clinicians systematically defer to model recommendation even when their own	H — Patients with atypical presentations (e.g. silent MI in diabetic women) are consistently undertriaged	Annual simulation-based training where staff must correctly override the model in ≥5 adversarial scenarios; pass rate required ≥90% before continued clinical access	Simulation pass rates logged annually; staff failing re-sit within 2 weeks or removed from triage duties pending retraining

			clinical judgment contradicts it	because clinicians do not override model errors		
10	Informed Consent Gap	Ethical	M — Patients presenting to Mercer General ED are not currently informed that an AI system contributes to their triage classification	M — Patients unaware of AI involvement cannot raise concerns or challenge decisions, breaching established ethical standards	Patient-facing disclosure statement developed in plain English and Creole, reviewed by a patient advocate group before deployment; displayed at triage desk and on admission paperwork	Patient advocate sign-off on file pre-deployment; disclosure statement reviewed annually or after any model update
11	Accountability Ambiguity	Ethical	M — No clearly named clinical owner exists for reviewing cases where the AI triage recommendation led to a poor outcome	H — When a patient is harmed following an AI-influenced triage decision, no clear governance pathway exists, exposing Mercer and the Ministry to legal and regulatory liability	Named Clinical AI Governance Owner (Dr. De Freitas or delegated consultant) documented in deployment SOP; case-review protocol activated within 24 hours of any adverse AI-influenced event	Case-review SOP exercised in at least two tabletop drills annually; zero adverse AI-influenced events without a completed case-review report within 5 working days
12	Training Data Underrepresentation	Equity	H — Caribbean ED populations include high proportions of elderly patients with NCDs, walk-in patients, and vector-borne illness presentations underrepresented in Mercer's historical data	H — Elderly women with atypical cardiac symptoms are systematically undertriaged at a higher rate than the overall population due to poor subgroup representation	Performance metrics disaggregated by age group, gender, arrival method, and primary complaint reviewed before each rollout; subgroup recall gap >5 pp triggers retraining before deployment	Subgroup performance report published quarterly; no deployment approved if any demographic subgroup recall lags overall recall by >5 pp
13	Language and Literacy Barriers	Equity	M — A significant proportion of Mercer General patients and some clinical staff communicate primarily in Trinidad Creole; English-only interface excludes this group	M — Triage staff misinterpret AI alerts or patients cannot engage with the system, creating inequitable access to AI-supported triage for non-English-dominant patients	Interface localisation completed in English and Trinidad Creole before go-live; plain language review conducted by a literacy specialist; translation reviewed annually	Localisation sign-off on file pre-deployment; patient satisfaction survey includes language accessibility item; score below 80% triggers interface review

Risk Memo:

Verdict

The proposed AI triage system at Mercer General carries manageable risk if three conditions are met: the model is validated against local patient patterns, clinicians retain independent judgement, and performance is checked separately for every demographic group.

Top 3 Risks

Risk 1 — The model may behave differently on patients it was not trained on

If a dengue outbreak, a carnival-season surge, or a rollout to another island brings patient presentations the model has never seen, its triage predictions may become unreliable without anyone noticing immediately.

Mitigation: The Clinical AI Lead re-evaluates model accuracy every month against a reserved set of unseen records, and any drop of more than 3 percentage points in correctly identifying the most critically ill patients triggers a mandatory clinical review before the model is used again.

Signal of success: A monthly accuracy report is published and on file; no drop of this size has gone unreviewed in the past quarter.

Risk 2 — Nurses may stop questioning the system, even when it is wrong

When a tool consistently gives confident recommendations, clinicians naturally begin to defer to it, and patients with unusual presentations, such as an elderly woman having a heart attack without typical chest pain, may wait too long for care because no one challenged the model's output.

Mitigation: The model output is displayed as a suggestion only, nurses must record their own assessment before seeing the recommendation, and all triage staff must pass an annual simulation exercise where they correctly override the model in at least five realistic test scenarios.

Signal of success: The weekly clinician override rate is tracked, and a sudden drop below 5% triggers a mandatory simulation drill and departmental review.

Risk 3 — The model may be less accurate for some patient groups than others

If elderly patients, walk-in patients, or patients presenting with vector-borne illnesses are underrepresented in Mercer General's historical records, the model may make systematically worse triage decisions for exactly these groups without the overall accuracy figures ever revealing the problem.

Mitigation: Before any deployment, model accuracy is broken down by age group, gender, mode of arrival, and primary complaint, and any group performing more than 5 percentage points below the overall average triggers retraining before go-live.

Signal of success : A subgroup performance report is published quarterly, and no deployment is approved if any demographic group falls below the 5 percentage point threshold.

Residual

At this stage, it is not yet possible to determine how Mercer General's triage nurses will respond to the system in real practice, whether they will trust it too much, ignore it, or integrate it appropriately, and this can only be assessed once the pilot begins, which is why the human override design and weekly monitoring are non-negotiable from day one.

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